

NON-LINEAR REGRESSION & BIOMETRIC PREDICTION

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*Dataset Domain: Biomechanics & Kinesiology
(bodyfat.csv)*

Target Variable: BodyFat (Continuous Percentage)

Assigned Algorithm: Decision Tree Regressor

1. Data Architecture & Data Leakage Prevention

The dataset consists of 252 physiological observations. The primary objective is to predict continuous body fat percentage based on anthropometric measurements (circumferences of the abdomen, chest, thigh, etc.).

The Siri Equation Trap (Data Leakage)

In physiological science, body fat percentage is not directly measured; it is mathematically derived from body density using Siri's Equation:

$$\text{BodyFat} = \frac{495}{\text{Density}} - 450$$

*Because the dataset includes a Density column, passing this feature to the machine learning model creates catastrophic **data leakage**. The model would simply learn to approximate Siri's algebraic formula rather than discovering physiological relationships across the actual body measurements.*

***Preprocessing Action:** The Density column was explicitly dropped prior to the train-test split to enforce true predictive learning.*

2. Mathematical Foundation of Decision Tree Regression

Unlike Ridge and Lasso regression (which apply continuous linear coefficients), a Decision Tree Regressor is a non-linear algorithm that uses recursive partitioning to split the dataset into distinct regions.

At each node, the algorithm evaluates every feature and every possible threshold to find the split that maximizes the reduction in variance, defined by the Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The algorithm calculates the variance of the parent node and subtracts the weighted variance of the proposed child nodes. The split that yields the largest variance reduction is selected. This process repeats until a stopping criterion is met.

3. Evaluation Metrics

To rigorously evaluate the regression models, three standard mathematical metrics were calculated:

- **Root Mean Squared Error (RMSE):** Measures the average magnitude of the error in the same units as the target variable (body fat percentage), heavily penalizing larger errors.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **Mean Absolute Error (MAE):** The average absolute distance between the predicted and actual values. It provides a linear penalty for errors.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **Coefficient of Determination (R^2):** Represents the proportion of variance in the dependent variable that is predictable from the independent variables.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

4. Model Training & Hyperparameter Optimization

To prevent the algorithm from memorizing the training data, a baseline model was compared against an optimized model pruned via `GridSearchCV`.

4.1 Baseline Decision Tree

A default decision tree expands nodes until all leaves are pure or contain less than two samples.

- **RMSE:** 5.5455
- **MAE:** 4.2706
- **R^2 Score:** 0.3389

The baseline model performed poorly on unseen testing data. By growing without limits, the tree over-segmented the continuous biological data, creating hyper-specific mathematical boundaries that failed to generalize.

4.2 Grid Search Optimization (Pruning)

To enforce regularization, a cross-validated grid search optimized the tree's structural limits, testing constraints on maximum depth, minimum samples required to split, minimum samples per leaf, and maximum features evaluated per split.

Optimal Parameters Discovered:

- **max_depth:** 3 (Restricts the tree to a maximum of 3 consecutive splits).
- **max_features:** 'sqrt' (Limits the algorithm to evaluate only the square root of total features at each node, preventing dominant features from masking secondary patterns).
- **min_samples_leaf:** 5 (Forces every terminal node to contain at least 5 patients).

- *min_samples_split*: 2

Optimized Performance:

- ***RMSE***: 4.8848
- ***MAE***: 4.0670
- ***R² Score***: 0.4870

Analytical Takeaway: Pruning the tree via mathematical constraints improved the R^2 variance explanation by approximately 43% (from 0.3389 to 0.4870) and reduced the absolute predictive error to $\approx 4\%$.

5. Feature Importance & Comparative Analysis

Decision Trees calculate feature importance by tracking the total reduction of MSE brought by that feature across all splits in the tree.

Top 5 Variance-Reducing Features:

1. ***Abdomen***: 0.8332 (83.3%)
2. ***Chest***: 0.0959 (9.6%)
3. ***Height***: 0.0242 (2.4%)
4. ***Knee***: 0.0238 (2.4%)
5. ***Hip***: 0.0121 (1.2%)

The algorithm determined that isolating the Abdomen circumference accounted for over 83% of the variance reduction in body fat prediction. The tree structurally ignored highly correlated appendages (like wrists, ankles, and forearms) in favor of core trunk measurements.

When comparing these non-linear results against linear Ridge or Lasso regression models (which penalize multicollinearity through L_2 and L_1 norms), the linear models will likely yield a higher R^2 . Biological morphology is highly continuous and correlated; Ridge and Lasso inherently distribute predictive weights across multiple features simultaneously. In contrast, this pruned Decision Tree demonstrates that a rigid, non-linear 3-step decision boundary focused predominantly on the abdomen remains highly capable of approximating physiological composition without complex multi-variable algebra.

6. Advanced Optimization: Cost-Complexity Pruning (CCP)

While *GridSearchCV* relies on discrete, pre-defined limits (such as strictly capping depth at 3), **Minimal Cost-Complexity Pruning** mathematically evaluates the exact variance reduction of every single node.

The algorithm introduces a complexity penalty (α) to the total Mean Squared Error of the tree. The cost-complexity measure $R_\alpha(T)$ of a tree T is defined as:

$$R_\alpha(T) = R(T) + \alpha|T|$$

Where $R(T)$ is the total training error of the tree, $|T|$ is the number of terminal leaf nodes, and α is the penalty applied to each additional leaf.

By calculating the entire pruning path, the algorithm dynamically collapses branches where the reduction in variance fails to overcome the α penalty.

6.1 CCP Results & Performance

The algorithm mathematically identified the optimal penalty parameter as $\alpha = 1.1579$. Applying this specific penalty forced the tree to naturally prune itself into a highly efficient structure with a maximum depth of 3 and exactly 6 terminal leaves.

Final Maximized Metrics:

- **RMSE:** 4.3389
- **MAE:** 3.5835
- **Final R^2 Score:** 0.5953

Analytical Takeaway: By allowing the math to dictate the pruning dynamically via the CCP algorithm, the model's R^2 score surged from 0.4870 (*Grid Search*) to **0.5953**. The model can now explain nearly 60% of the variance in human body fat using only a handful of non-linear decision rules.

7. Visual Model Interpretation (The Decision Boundaries)

Because a Decision Tree is a "white-box" model, the generated topology allows for exact physiological interpretation of the algorithm's logic.

1. The Root Split (Primary Variance Reducer):

The most mathematically significant physiological boundary is an **Abdomen circumference of 91.95 cm**. The algorithm determined that patients below this threshold (True branch) average a body fat of 13.49%, while those above it (False branch) average 25.81%.

2. The High-Risk Sub-Group (Right Branch):

For individuals with an abdomen > 91.95 cm, the tree continues to split entirely on the Abdomen feature (at thresholds of 103.0 cm and 112.3 cm). The terminal leaf on the far right identifies the highest body fat cluster (averaging 35.65% body fat) specifically for individuals with an abdomen > 112.3 cm.

3. The Low-Risk Sub-Group (Left Branch):

For individuals with an abdomen ≤ 91.95 cm, the algorithm introduces **Weight** as the secondary physiological determinant. For individuals with an abdomen ≤ 85.45 cm, if their weight is ≤ 173.62 lbs, they fall into the lowest possible body fat cluster (averaging 9.95%).

Final Conclusion: The physical topology of the tree definitively proves that while linear models (Ridge/Lasso) may assign small coefficients to the neck, thigh, or biceps, the non-linear Decision Tree successfully approximates physiological composition using strictly core trunk mass (Abdomen) and overall mass (Weight).

